# SIMSHIELD

*Network-native identity verification & continuous KYC fraud prevention*

GSMA Open Gateway APIs · Nokia NaC Platform · Google Gemini AI

*Open Gateway Hackathon 2026 · Talent Arena*

# SIMSHIELD

*Network-native identity verification*
*& continuous KYC fraud prevention*

**Open Gateway Hackathon 2026 · Talent Arena**

Nokia Network as Code · GSMA Open Gateway

# "Every fraud attack starts with a SIM card nobody checked."

SIMShield makes the telecom network
the identity layer it was always meant to be.

| | | | |
|---|---|---|---|
| **SIM Swap** | **KYC Match** | **Device Swap** | **Device Location** |

4 LIVE APIs · Nokia NaC · Movistar · Orange · Vodafone

| | | | |
|---|---|---|---|
| **< 300ms** | **DORA Ready** | **Zero-Data** | **3 Operators** |
| avg latency | EU 2025 | GDPR-native | ~95% Spain |

Presented by: Michella (Product Owner) · Victor (AI/Tech) · Armando (PoC/Dev)

# THE PROBLEM — IDENTITY FRAUD IS A SYSTEMIC RISK. CURRENT DEFENCES WERE NOT BUILT FOR THIS.

*Business Relevance: Real persona · Concrete consequences · Why acting now protects operators and clients*

## THE SCENARIO

**María is shopping in Madrid. 450km away in Seville, a fraudster tries to use her cloned card at a POS terminal.**

**The bank fires a verification request to SIMShield. We check via Device Location API: SIM swapped 38 minutes ago. Device in Seville. Card registered in Madrid. Score: 9.**

**SIMShield's chatbot alerts María: "Are you in Seville?" She replies: "No."**

**Transaction denied. Fraud stopped.**

**£300M**

**M&S fraud loss**

Outsourced helpdesk SIM swap
Q1 2025 — in this room's sector

**1%**

**SIM swap surge**

UK 2023→2025. Spain following.
Fraud ring industrial scale.

**€2.1B**

**Card-present EU fraud**

POS terminal attacks.
Phone 450km from card.

## WHY CURRENT DEFENCES FAIL

### SMS 2FA

Bypassed when swap completes.
OTP routes to attacker.
Bank has zero visibility.

### Bank Fraud AI

Fires after transaction.
SIM swap happened minutes before — outside bank window.

### Static KYC

Checked once at onboarding.
Never re-verified when SIM changes hands mid-relationship.

### Carrier PINs

Outsourced helpdesk bypass under pressure. M&S exact vector. Not scalable.

**FOR OPERATORS — PROACTIVE FRAMING:** *Reducing fraud proactively lowers chargeback liability, reduces regulatory pressure, and protects operator brand — before AEPD enforcement. Fine exposure and regulatory detail are out of scope today but have been modelled — ask us in Q&A.*

# PoC ARCHITECTURE — 4 LIVE APIs · NOKIA NaC · sim-shield.vercel.app · < 300ms

*Technical Demonstration: Live on Vercel · Nokia NaC platform · 3 operators · GSMA CAMARA standard*

**SIMShield** verifies the SIM, not just the password — before every transaction, in real time, on all 3 Spanish operators.

### USER LAYER

Merchant POS · Bank App · Enterprise Helpdesk · ADP HR Platform

### SIMSHIELD PRODUCT LAYER

FastAPI backend · Gemini AI Agent · MCP Protocol · Vercel deployment · React dashboard + Mapbox

### NOKIA NaC PLATFORM

network-as-code.developer.nokia.com · GSMA CAMARA s

**SIM Swap**
Swapped last 24h?

or

**KYC Match**
Name/DOB vs live record

**Device Swap**
Hardware changed?

**Device Location** • LIVE
Phone at POS? Delta calc.

### OPERATOR NETWORK LAYER

Movistar · Orange · Vodafone — 3 live 5G networks — ~95% Spanish mobile subscribers

### TECH STACK

**Backend:** Python FastAPI **Deploy:** Vercel **APIs:** Nokia NaC SDK **AI:** Gemini + MCP **Frontend:** React + Mapbox **Pro**

• sim-shield.vercel.app

**287ms**
avg latency per check

# AI INTEGRATION + SECURITY & COMPLIANCE — GEMINI · MCP · ZERO-DATA BROKER · DORA READY

*Technical Innovation: Agentic AI for real decisions · Human-in-loop · GDPR-native · Financial risk reduction*

## MCP TOOLS

- `check_sim_swap()`
- `kyc_match()`
- `device_swap()`
- `check_location()`

## DECISION CHAIN

**TRIGGER**
Payment · Login · Helpdesk

**02 PARALLEL CALLS**
4 MCP tools via Nokia NaC

**03 AI REASONING**
Gemini chain-of-thought

**04 SCORE 0–100**
SIM 30% · KYC 25% · Loc 20% · Dev 15%

**05 DECIDE / A2A**
<30 approve · 30-70 escalate · >70 block

## GEMINI VERDICT — DEMO MOMENT

*"SIM swapped 38 min ago.*
*Device 450km from POS.*
*KYC mismatch confirmed.*
*Confidence: 97%.*

*Recommendation: BLOCK."*

**BLOCKED — 97% CONFIDENCE**

## SECURITY & GDPR

- Zero raw data on SIMShield servers — hashed tokens only
- Gemini sees signals only — never MSISDN, name, coordinates
- TLS 1.3 · Cloud Secret Manager · GDPR Art. 6(1)(f)

## COMPLIANCE & FINANCIAL RISK REDUCTION — WHY OPERATORS & MNCs ACT NOW

### DORA (EU 2025)

Real-time identity controls mandated. SIMShield is compliant by design.

### PSD3 / SCA

Stronger auth required. SIM verification is the missing layer.

### AEPD Awareness

Operator fine risk noted. Out of scope today — ask us in Q&A.

### MNC Risk (ADP)

Helpdesk impersonation = payroll fraud vector. M&S blueprint.

### Operator ROI

Lower chargebacks, lower regulatory risk, 15% rev-share.

# COMMERCIAL VIABILITY — PHASED FINANCIAL MODEL · CLIENT TIERS · ROI

*Commercial Viability: Business model · Target market · Value proposition · Phase 1 MVP → Phase 2 MVT → Phase 3 Scale*

## TARGET CLIENT SEGMENTS

**BEACHHEAD — MVP**

### Spanish Neobanks & Fintechs

Rebellion Pay · Bnext · Revolut ES · N26 ES

**€0.004/call**

**SCALE — PROCESSORS**

### Payment Processors

Redsys — 1 contract = national coverage

**Rev share**

**ENTERPRISE MNC — MVT**

### Large Corporates + HR Platforms

ADP · Workday · SAP SuccessFactors · BBVA · Santander

**€8/seat/mo**

**SME — HOLDED/QUIPU**

### Spanish SMEs via Plugin

200K addressable · Holded + Quipu ecosystem

**€29/mo**

| TAM | SAM | SOM |
|---|---|---|
| $47B | $8.4B | $220M |

## PHASED FINANCIAL MODEL

| | PHASE 1<br>MVP<br>Now → Q2 2026 | PHASE 2<br>MVT Launch<br>Q3 2026→Q1 2027 | PHASE 3<br>Scale<br>Q2→Q4 2027 |
|---|---|---|---|
| Scope | PoC · demo day | | 20 fintechs + 2 MNC pilots |
| Monthly checks | Live demo | | ~2M / month |
| ARR | €0 | | ~€85K |
| Cost / check | €0.036 avg | | €0.036 avg |
| Break-even | — | | Month 8–10 |
| MNO rev share | — | | 15% to operators |
| Spain API $$ | Not monetised | | Launching 2026 |
| Key milestone | Hackathon win | | Redsys integration |

## NOKIA + TELEFÓNICA TESTBED vs SIMSHIELD — 4 SPECIFIC GAPS

| Dimension | Nokia + Telefónica (Lab) | SIMShield (Product) |
|---|---|---|
| APIs used | SIM Swap + Device Swap (2 APIs) | SIM Swap + KYC Match + Device Swap + Device Location (4 APIs) |
| Operators | Telefónica only — single MNO | Movistar + Orange + Vodafone — ~95% Spain |
| KYC signals | None | KYC Match: live name/DOB vs operator record |
| Geolocation | None — digital fraud only | Device Location vs POS — card-present fraud stopped |
| AI layer | A2A + MCP protocol testing | Gemini making real decisions · scored output · human-in-loop |
| Commercial | Lab PoC, no pricing | €0.004/call · €29/mo SME · €8/seat MNC · deployable today |
| Compliance | Not addressed | DORA-ready · zero-data broker · GDPR Art. 6(1)(f) |

## SCALABILITY

- GSMA CAMARA → any operator globally
- Nokia NaC SDK → single integration, multi-MNO
- Cloud-native stateless API → 50M checks/mo
- ADP/Workday → 1 HR integration = all employees

## ROI SUMMARY

| | |
|---|---|
| 1 incident prevented pays for: | > 116,000 checks |
| Cost per check: | ~€0.036 incl. SMS |
| ROI per prevention: | > 2,000× |

**NOKIA BUILT THE PIPES. SIMSHIELD BUILT THE PRODUCT. THE APIS ARE LIVE. THE OPERATORS ARE IN THIS ROOM.**

*That product is SIMShield.*

SIMSHIELD

outline

SLIDE 1
PROBLEM STATEMENT – FRAUD IN NUMBERS
PROBLEM STATEMENT
SLIDE 2 –
ONE LINE SOLUTION: **REAL-TIME NETWORK VERIFICATION USING MOBILE OPERATOR APIs BEFORE FUNDS MOVE**
**USER FLOW**

- **BUILT ON NaC platfrm**
- **Global system for mobile Association GSMA Open gateway framework**

SLIDE 3 – Value
WHY COMBINING 5 APIs CREATES VALUE that one API alone cannot
SLIDE 4- Agentic AI MAKES DECISIONS, NOT DECORATIONS

- OUR SYSEM USES GEMINI THROUGH MODEL CONTEXT PROTOCAL (MCP) TO : DECIDE WHICH APIs TO CALL BSED ON TRANSACTIONAL RISK
- APPLE THRESHOLD LOGIC (ONLY TRANSACTIONS ABOVE 200€ TRIGGER FULL CHECK)
- RISK REDUCTION MODEL, NOT growth at all costs model; ours is where one ‘prevented incident’ funds tens of thousands of verifications
-

# COMPETITIVE LANDSCAPE



**The combination of real-time SIM verification + GSMA KYC APIs + AI reasoning does not exist as a product today.**

| | Real-time SIM Layer | GSMA KYC APIs | AI Reasoning Verdict | Spain/GSMA Native | Operator RevShare | SME Pricing |
|---|---|---|---|---|---|---|
| **SIMShield** | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Vonage Verify | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| Twilio Verify | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| Experian / LexisNexis | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Okta / Auth0 | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Bank internal KYC | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |

**White space: nobody combines all six. That's the moat.**

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# SIMSHIELD

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**SIMShield makes the telecom network the identity layer it was always meant to be. Every day, banks approve transactions based on a PIN and a password. What they don't check is whether the phone number on that account still belongs to the same person. We built SIMShield to close that gap.**

SIMShield makes the telecom network the identity layer it was always meant to be.

# Third-party system fraud protection

SIMSHIELD

*Network-native identity verification & continuous KYC fraud prevention*

## Our API conectors

**4 APIs · 3 operators · <300ms fraud verdict**

- SIM Swap
- KYC Match
- Device Location
- Device Swap

# SIMSHIELD

**Differentiator 1**
**KYC signal stack** Nokia/Telefónica testbed uses SIM Swap and Device Swap. You use 8 APIs, including the full KYC family: Match, Tenure, Age, and Fill-in. KYC Tenure specifically — how long has the SIM been active — is not in their testbed and is one of the strongest fraud signals available. A 2-day-old SIM on a 10-year-old bank account is a fraud ring signature. Their system cannot see that. Yours can.

**Differentiator 2**
**Multi-operator aggregation** Their testbed is Telefónica only. SIMShield runs across Movistar, Orange and Vodafone simultaneously. In Spain, that means ~95% of mobile subscribers are covered by one API call. A fraud check that works only with one operator is not commercially viable. Nokia NaC is the aggregation layer that makes this possible

**Differentiator 3**
**Physical payment geolocation** Their use case is purely digital fraud — SIM swap for account takeover. SIMShield adds the physical payment layer: Location Retrieval API compared against POS terminal coordinates. This is a completely new use case that their testbed does not cover. Card-present fraud at physical terminals is a €2.1B annual problem in Europe that no network-layer product addresses today.

**Differentiator 4**
**Commercial product with SME pricing , Nokia** and Telefónica are building infrastructure and protocols. They are not building a €29/month Holded plugin for Spanish SMEs. They are not building a DORA compliance dashboard for bank KYC officers. They are not building an enterprise helpdesk verification layer. The commercial product and go-to-market is entirely yours. This is the feasibility argument — you have a path to revenue that their initiative explicitly does not pursue.

# THE PROBLEM



*Static KYC has a blind spot: the SIM card*

**£300M**

M&S lost Q1 2025 to SIM swap social engineering through an outsourced IT helpdesk. No zero-day. A phone call.

**1,055%**

Surge in unauthorised UK SIM swaps in 2024. eSIM self-service cut attack time from hours to under 5 minutes.

**€5.8M**

AEPD fines on Vodafone ES, Orange, Telefónica for failing identity checks at SIM duplication. Already happened.

**Root cause: KYC is checked once at onboarding. Never again at transaction time. The SIM can change hands — the bank never knows.**

## WHY EVERY CURRENT DEFENCE FAILS

- SMS 2FA: Intercepted the moment swap completes. OTP goes to attacker's device, not yours.
- Static KYC: Checked once at onboarding. Never re-verified when the SIM changes hands.
- Bank fraud AI: Fires after the transaction. Swap happened minutes earlier, outside bank visibility.
- Carrier PINs: Outsourced helpdesk agents bypass under time pressure via scripted social engineering.

# KNOW YOUR CUSTOMER — AND THE GAP GSMA CLOSES



## STANDARD KYC TODAY

*Checked once at account opening.*
*Never again.*

- ✓ Name match
- ✓ Date of birth
- ✓ Address verification
- ✓ Document scan (DNI/Passport)
- ✗ SIM ownership not checked
- ✗ No re-verification at transaction
- ✗ No tenure signal
- ✗ No call forwarding check

## GSMA OPEN GATEWAY KYC APIs

*Live operator data. Checked at every transaction.*

### KYC Match

Does name/DOB/address match operator's live record for this number right now?

### KYC Tenure

How long has this SIM been active? 2-day SIM on 10-year account = fraud ring signature.

### KYC Age

Operator-verified age for regulated products — no document upload required.

### KYC Fill-in

Operator pre-fills verified personal data — reduces friction AND verifies accuracy.

**This is continuous KYC — not just at onboarding, at every transaction that matters.**

# THE SOLUTION — 6 APIS, ONE FRAUD VERDICT



**SIMShield makes the telecom network the identity layer it was always meant to be.**

| API | SIGNAL | FRAUD INDICATOR |
|---|---|---|
| **SIM Swap** | Was SIM changed recently? | Swap within 24h of transaction = high risk |
| **Number Verification** | Device cryptographically match number? | Mismatch = SIM hijacked, OTPs intercepted |
| **KYC Match** | Identity match operator live record? | Name/DOB mismatch = stolen identity in use |
| **KYC Tenure** | How long has this SIM been active? | <7 days on old account = fraud ring signature |
| **Device Status** | Roaming? Unexpected network? | Sudden roaming on domestic transaction = anomaly |
| **Call Forwarding** | Is OTP call forwarding active? | Forwarding to unknown number = active interception |

MODULE 1

### Transaction Guard

Silent 6-API check before any payment >€500.
Verdict in <300ms. Zero friction for legitimate users.

MODULE 2

### Continuous KYC

Replace static onboarding KYC with live re-verification on every login, beneficiary add, limit increase.

MODULE 3

### Zero-Trust Helpdesk

Verify caller's SIM matches corporate device before any password reset. Blocks the exact M&S attack.

# AI LAYER — MCP SERVER + GEMINI AGENT

*Built in, not bolted on*

**1**

**TRIGGER**

Payment >€500
New beneficiary
Helpdesk call

**2**

**MCP SERVER**

6 tools exposed:
check_sim_swap()
kyc_match() etc.

**3**

**GEMINI AGENT**

Calls tools
Reasoning chain
Verdict + explanation

**4**

**FRAUD SCORE**

0–100 weighted
confidence score
per signal

**5**

**ACTION**

Allow / Challenge
/ Block + Alert
via webhook

**GEMINI AGENT OUTPUT — DEMO MOMENT**

*"SIM was swapped 47 minutes ago. Device is now roaming on an unfamiliar Spanish network. KYC name match failed — operator record shows a different surname to the bank account. Call forwarding is active to an unregistered number. Recommendation: Block transaction and flag account for manual review. Fraud confidence: 97%."*

*Why MCP matters: SIMShield tools become callable by any AI agent — Gemini, Claude, or enterprise SOC tools. This is how security teams actually consume fraud intelligence.*

# WHY NOW — FOUR FORCES CONVERGING IN 2025



### AEPD ALREADY FINED THE OPERATORS IN THIS ROOM

Vodafone ES, Orange, Telefónica paid €5.8M for SIM identity failures. SIMShield is a compliance product for them — not a nice-to-have. The procurement conversation is already half won.

### GSMA OPEN GATEWAY IS LIVE ACROSS ALL THREE OPERATORS HERE

Not a future state. These APIs are running on Movistar, Orange and Vodafone today. You can demo this on a real SIM card in the next 24 hours.

### EU DORA + AI ACT MANDATE CONTINUOUS IDENTITY VERIFICATION

Financial institutions must demonstrate real-time identity controls from 2025. Static KYC is legally insufficient. SIMShield becomes compliance infrastructure, not a product choice.

### NO ONE HAS BUILT THE INTELLIGENCE LAYER YET

The APIs are live. The standard is set. The window is 12–18 months before Experian or LexisNexis builds this internally. First-mover on GSMA KYC APIs = durable moat.

*The operators in this room have regulatory, commercial and technical reasons to want this built. That's the rarest possible alignment.*

# MARKET SIZE



### TAM

**$47B**

Global financial fraud losses addressable by identity verification (CFCA 2024)

### SAM

**$8.4B**

European regtech / KYC compliance market, growing 23% YoY driven by DORA and EU AI Act

### SOM

**$220M**

Spanish financial institutions + enterprise IT helpdesks, 3-year capture at 12% penetration

## BEACHHEAD CUSTOMER

Spanish neobanks and fintechs — Revolut ES, N26, Rebellion Pay, Bnext.
Lighter compliance infrastructure. Higher fraud exposure. Faster procurement than traditional banks.
First sale achievable in under 90 days post-hackathon.

**Operator revenue share: 15% of API call revenue back to Movistar / Orange / Vodafone — aligns incentives with the operators in this room.**

# TECHNICAL ARCHITECTURE



*The technical contract for your engineers*

**1**
**TRIGGER**

Payment >€500
New beneficiary
Login new device
Helpdesk call

**2**
**PARALLEL APIs**

6 Nokia NaC API calls fired simultaneously
(SIM Swap + Verify + KYC x3 + Status + Fwd)
All in one round-trip <300ms

**3**
**SCORE ENGINE**

Weighted fraud score:
Swap recency 30%
KYC match 25%
Tenure 25%
Device anomaly 20%

**4**
**GEMINI AGENT**

MCP tools called
Chain-of-thought
Plain-English verdict
via Google Cloud

**5**
**ACTION**

Score <30 → Allow
Score 30–70 → Challenge
Score >70 → Block
+ Webhook alert

## TECH STACK

- Backend: Python FastAPI + Nokia NaC SDK
- AI Agent: Google Gemini via Google Cloud (free access provided)
- MCP: Python MCP SDK — fraud engine exposed as 6 tools
- Frontend: React + Recharts — live fraud event feed + KYC status panel
- Demo: Live SIM card on Movistar / Orange / Vodafone 5G

**Uses at least 6 GSMA APIs across all 3 operators (Movistar, Orange, Vodafone) — maximum Technical Innovation score.**

# BUSINESS MODEL



### TIER 1
### FINTECH API

**€0.004**
**per verification**

→ Revolut ES, N26, Rebellion Pay

→ 100M+ verifications/yr at scale

→ Fraud prevention ROI >1,000×

→ No subscription lock-in

### TIER 2
### CONTINUOUS KYC

**€8**
**seat / month**

→ DORA compliance dashboard

→ Replaces manual re-verification

→ Santander, BBVA, CaixaBank

→ Regulator audit trail built in

### TIER 3
### SME GUARD

**€29**
**month**

→ Holded / Quipu plugin

→ ~200K addressable Spanish SMEs

→ Avg prevented loss: €8,000

→ ROI: one prevented incident = payback

Year 1 target: 200 fintechs + 5 enterprise + 300 SMEs = ~€310K ARR · Break-even: 14 months · 15% operator revenue share aligns Movistar/Orange/Vodafone as channel partners

# TEAM & SPRINT PLAN



*BCG X engagement model — PM-led, outcome-focused*

### PM / PRODUCT OWNER — *You*

Owns pitch, demo narrative, scope. Runs kickoff agenda. Cuts features ruthlessly. The M&S/AEPD story is yours to tell to the jury.

### BACKEND ENGINEER — *TBD*

FastAPI. 6 Nokia NaC API integrations. MCP server implementation. Priority: SIM Swap → Number Verify → KYC Match → KYC Tenure → Device Status → Call Forwarding.

### ML / DATA ENGINEER — *TBD*

Fraud score model. Features: tenure weight 25%, swap recency 30%, KYC match 25%, device anomaly 20%. Must run on synthetic data before frontend starts.

### FRONTEND ENGINEER — *TBD*

React dashboard. Real-time fraud feed. KYC status panel. Gemini verdict display. Owns the visual layer judges see.

## SPRINT TIMELINE

| | |
|---|---|
| **MON MORNING** | Org switch · Google Cloud · SIM cards · Kickoff agenda |
| **MON AFTERNOON** | Backend: SIM Swap + Number Verify + KYC Match API calls live |
| **MON EVENING** | ML: fraud model running · Frontend: skeleton renders |
| **TUE 06:00–09:00** | MCP server + Gemini agent · Live SIM demo confirmed |
| **TUE 09:00–10:30** | Full demo rehearsal · Pitch under 5 min · Scope cuts if needed |
| **TUE 11:00** | SUBMISSION DEADLINE |

# PM EXECUTION CHECKLIST — SCOPE & TEAM KICKOFF



## KICKOFF AGENDA (60 MIN)

| | |
|---|---|
| 0:00–0:10 | Problem alignment — slides 2–3. Does everyone agree on the core problem? |
| 0:10–0:20 | Scope cut — Module 1 (Transaction Guard) ships first. Non-negotiable. |
| 0:20–0:30 | Role assignment — Backend, ML, Frontend confirmed. No overlap. |
| 0:30–0:45 | API priority order agreed — SIM Swap first, KYC Tenure last. |
| 0:45–0:55 | Sprint 1 backlog — 3 user stories per role. PM prioritises live. |
| 0:55–1:00 | Standup cadence — 9am daily, 10:30pm demo check-in Tuesday. |

## GO / NO-GO — NOTHING GETS BUILT UNTIL TICKED

- ✓ Nokia NaC org switched to team organisation (dashboard dropdown)
- ✓ Google Cloud account activated — Gemini API key obtained
- ○ 5G SIM card obtained from organisers (Movistar/Orange/Vodafone)
- ✓ SIM Swap API returning live event confirmed (<200ms)
- ○ KYC Match API returning real operator data confirmed
- ○ KYC Tenure API returning SIM age confirmed
- ✓ Fraud score model running on synthetic dataset
- ○ MCP server exposing at least 3 tools to Gemini agent
- ○ React dashboard renders with live API data — no mocks
- ○ Demo script: problem → 6 signals → Gemini verdict → block
- ○ Pitch rehearsed at full speed — confirmed under 5 minutes